INCH-POUND

TT-T-548F
November 1, 1991
SUPERSEDING
TT-T-548E
April 22, 1986

FEDERAL SPECIFICATION

TOLUENE, TECHNICAL

This specification is approved by the Commissioner of Federal Supply Service, General Services Administration, for use by all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one technical grade of toluene.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-C-96 – Cans, Metal, 28 Gauge and Lighter

Federal Standards:

FED. TEST METHOD STD. NO. 101 – Test Procedures for Packaging Materials
Fed. Std. No. 123 – Marking for Shipment (Civil Agencies)
FED-STD-313 – Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to the Government

Comments or suggestions pertaining to this specification should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423.

AMSC N/A FSC 6810

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(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Washington, DC; Philadelphia, PA, Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Auburn, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-P-15011 – Pallets, Material Handling, Wood Post Construction, 4-Way Entry

Military Standards:

MIL-STD-129 – Marking for Shipment and Storage
MIL-STD-147 – Palletized Unit Loads

(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Code of Federal Regulations (CFR):

29 CFR 1900.1200 – Hazard Communication
49 CFR 171 to 199 – Hazardous Materials Regulations

(The Code of Federal Regulations and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM Standards:

C 516	–	Vermiculite Loose Fill Thermal Insulation
D 362	–	Industrial Grade Toluene
D 847	–	Acidity of Benzene, Toluene, Xylenes, Solvent Naphthas, and Similar Industrial Aromatic Hydrocarbons
D 848	–	Acid Wash Color of Industrial Aromatic Hydrocarbons
D 849	–	Copper Corrosion of Industrial Aromatic Hydrocarbons
D 850	–	Distillation of Industrial Aromatic Hydrocarbons and Related Materials
D 853	–	Hydrogen Sulfide and Sulfur Dioxide Content (Qualitative) of Industrial Aromatic Hydrocarbons
D 891	–	Specific Gravity, Apparent, of Liquid Industrial Chemicals
D 1209	–	Color of Clear Liquids (Platinum–Cobalt Scale)
D 1296	–	Odor of Volatile Solvents and Diluents
E 29	–	Using Significant Digits in Test Data to Determine Conformance with Specifications

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

International Civil Aviation Organization

"Technical Instructions for the Safe Transport of Dangerous Goods by Air"

(Application for copies should be addressed to the International Civil Aviation Organization, 1000 Sherbrooke Street West, Suite 400, Montreal, Quebec, Canada H3A 2R2.)

International Maritime Organization

"International Maritime Dangerous Goods Code"

(Application for copies should be addressed to the International Maritime Organization, 101–104 Piccadilly, London, WIV OAE, England.)

(Non–Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. Toluene shall conform to the chemical and physical characteristics specified herein when tested as specified in the applicable portions of 4.2.4.

3.1.1 Specific gravity. Toluene shall have a specific gravity at 20°/20°C of no less than 0.860 and no more than 0.870.

3.1.2 Color. Toluene shall have a color no darker than No. 20 on the platinum-cobalt scale.

3.1.3 Distillation range. Toluene shall have a distillation range of no more than 2°C at 760 millimeters of mercury pressure from the initial boiling point to the dry point. The distillation range shall include the temperature of 110.6°C.

3.1.4 Residual odor. Toluene shall have no residual odor.

3.1.5 Water. Toluene shall have no evidence of turbidity or free water at 20°C.

3.1.6 Acidity. Toluene shall contain no free acid (no evidence of acidity).

3.1.7 Acid wash color. Toluene shall have an acid wash color no darker than a No. 4 color standard.

3.1.8 Sulfur compounds. Toluene shall be free of hydrogen sulfide and sulfur dioxide.

3.1.9 Copper corrosion. Toluene shall pass the copper corrosion test.

3.2 Precautionary marking. Precautionary marking shall be as follows:

HAZARDS:
Flammable Liquid
Toxic by Inhalation, Ingestion, and Absorption

WARNING! FLAMMABLE
HARMFUL IF INHALED, SWALLOWED, OR ABSORBED THROUGH SKIN

Keep away from heat, sparks, and open flame.
Keep container closed.

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Use with adequate ventilation.
Avoid breathing vapor.
Avoid contact with eyes, skin, and clothing.
Wash thoroughly after handling.

FIRST AID: Remove contaminated clothing and shoes. Flush affected areas with plenty of water. If in eyes, hold eyelids open and flush with plenty of water. If swallowed and the victim is conscious, have victim drink water or milk. Do not induce vomiting. If inhaled, move to fresh air. If breathing difficult, give oxygen. If in eyes, swallowed, or inhaled, get medical aid.

In case of fire, extinguish with dry chemical, foam, or carbon dioxide.

3.3 Material Safety Data Sheets. Material Safety Data Sheets for toluene shall be prepared and submitted by the contractor in accordance with FED-STD-313 (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling

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procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.4.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the toluene produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.5).

4.2.2 Sampling. See 6.6 for sampling and testing precautions.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with table I. The sample unit shall be one filled unit pack or packing container, as applicable, ready for shipment.

TABLE I. **Sampling for packaging examination and test**

Number of containers in batch or lot	Number of sample containers
1 or 2	all
3 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1,200	15
1,201 to 3,200	18
3,201 to 10,000	22
over 10,000	29

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4.2.2.2 For toluene tests. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 0.9 liter shall be removed from each sample container and placed in a suitable clean, dry closed container labeled to identify the lot and container from which it was taken.

TABLE II. **Sampling for toluene tests**

Number of containers in batch or lot	Number of sample containers
1 or 2	all
3 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage tests. Sampling shall be conducted in accordance with table I. The sample unit shall be one filled and closed container.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. Sample containers shall be examined for the characteristics listed below. Failure of any sample container to conform to all characteristics shall be cause for rejection of the lot represented.

(a) Contents per container
(b) Container
(c) Container closure
(d) Container free of damage and leaks
(e) Container coating
(f) Fiberboard packing aids
(g) Wood preservation evident and correct (level A boxes only)
(h) Marking evident, correct and legible
(i) Unitization (when required)

4.2.3.2 For toluene tests. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5. Failure of the leakage test by any container shall be cause for rejection of the lot represented.

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4.2.3.4 Significant places. For the purpose of determining conformance with this specification, and observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with ASTM E 29.

4.2.4 Toluene tests. See 6.6 for sampling and testing precautions. The requirements listed in table III shall be determined in accordance with the applicable ASTM test methods specified therein.

TABLE III. **Toluene test methods**

Requirement	Requirement paragraph	ASTM test method
Specific gravity	3.1.1	D 891*
Color	3.1.2	D 1209
Distillation range	3.1.3	D 850**
Residual color	3.1.4	D 1296
Water	3.1.5	D 362
Acidity	3.1.6	D 847
Acid wash color	3.1.7	D 848
Sulfur compounds	3.1.8	D 853
Copper corrosion	3.1.9	D 849

* Measured to the third decimal place
**Using an ASTM No. 41C thermometer

4.2.5 Container leakage tests.

4.2.5.1 One-quart and one-gallon unit quantities. Filled and closed 1-quart and 1-gallon unit quantity containers shall be tested for leakage in accordance with method 5009.3 of FED. TEST METHOD STD. NO. 101, using the hot water technique for leakage of air. A steady stream of bubbles or toluene globules shall be considered evidence of container leakage.

4.2.5.2 Five-gallon and fifty-five-gallon unit quantities. Filled and closed 5-gallon and 55-gallon unit quantity containers shall be tested for leakage in accordance with method 5009.3 of FED. TEST METHOD STD. NO. 101, using the static leak test for leakage of water or other contents.

5. PACKAGING

Note: The metric equivalents given for inch-pound quantities are nominal values provided for informational purposes and should not be considered as quantity requirements.

5.1 Packaging. Packaging shall be in accordance with the applicable requirements of 49 CFR 171 to 199 and either the International Civil Aviation Organization - Technical Instructions for the Safe Transportation of Dangerous Goods by Air (ICAO-TDGA) or

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International Maritime Organization – International Maritime Dangerous Goods Code (IMO-IMDGC), as applicable to the mode of transportation. The packaging shall meet the applicable packaging performance tests specified in ICAO-TDGA or IMO-IMDGC, as applicable.

5.2 Unit packing.

5.2.1 One-quart (1-liter) quantity. A quantity of 1 quart (+1/4 or -0 fluid ounces) (1 liter) of toluene shall be unit packed in a nominal 1-quart (1-liter) oblong can conforming to an IP.3 container of ICAO-TDGA or IMO-IMDGC, as applicable. The can shall conform to type V, class 4, with plan B coating with side seam striping of PPP-C-96. The can and closure shall be formed from commercial designation minimum No. 100 electrolytic tinplate. The cap shall have a phenolic-faced pulpboard liner. The closure shall be torqued to 20 (+0 or -5) inch-pounds. There shall be no evidence of leakage when the filled and closed container is tested as specified in 4.2.5.1.

5.2.2 One-gallon (4-liter) quantity. A quantity of one gallon (+1 or -0 fluid ounces) (4 liters) of toluene shall be unit packed in a nominal 1-gallon (4-liter) can conforming to the requirements specified in 5.2.1, except that the can shall be furnished with a bridgetype handle. In addition the can shall have a leak-proof, tin-plated, press-fitted inner seal. The closure shall be torqued to 35 (+0 or -5) inch-pounds. There shall be no evidence of leakage when the filled and closed container is tested as specified in 4.2.5.1.

5.3 Packing. Packing shall be level A or B as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 One-quart (1-liter) quantity. Twenty-four 1-quart (1-liter) cans of toluene packaged as specified in 5.2.1 shall be packed upright in two tiers with each tier having a three by four can arrangement in a close-fitting, water-repellent treated wooden or plywood box. The wooden box shall conform to the requirements of a 4C1 container of ICAO-TDGA or IMO-IMDGC, as applicable. The plywood box shall conform to the requirements of 4D container of ICAO-TDGA or IMO-IMDGC, as applicable. In addition, the box shall have a 4 mil Polyethylene bag of sufficient size to form a liner and a sufficient quantity of vermiculite, conforming to type I, grade 3 of ASTM C 516, to surround the cans to assure absorption of the entire contents in the event of leakage or breakage. Also the plywood shall be not less than 3/8 inch minimum thickness and the cleats shall be not less than 3/4 inch thick by 1-3/4 inch wide minimum. Prior to closing the box, the bag shall be heat sealed. Each box shall be closed and reinforced as specified in accordance with the general packing requirements of ICAO-TDGA or IMO-IMDGC, as applicable.

5.3.1.2 One-gallon (4-liter) quantity. Six 1-gallon (4-liter) cans of toluene shall be packed as specified in 5.3.1.1, except that the pattern shall be two by three.

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5.3.1.3 Five-gallon (20-liter) quantity. A quantity of five gallons (+5 or -0 fluid ounces) (20 liters) of toluene shall be packed in tight-head metal pail conforming to the requirements of a 1A1 container of ICAO-TDGA or IMO-IMDGC, as applicable. There shall be no evidence of leakage when the filled and closed container is tested as specified in 4.2.5.2.

5.3.1.4 Fifty-five-gallon (220-liter) quantity. A quantity of 55 gallons (+1/2 or -0 gallons) (220 liters) of toluene shall be packed in a steel drum conforming to the requirements of a 1A1 container of ICAO-TDGA or IMO-IMDGC, as applicable to the mode of transportation. There shall be no evidence of leakage when the filled and closed container is tested as specified in 4.2.5.2.

5.3.2 Level B

5.3.2.1 One-quart (1-liter) quantity. Twenty-four 1-quart (1-liter) cans of toluene unit packed as specified in 5.2.1 shall be packed upright as specified in 5.3.1.1 except that the container shall be a close-fitting, weather-resistant fiberboard box conforming to the requirements of a 4G container of ICAO-TDGA or IMO-IMDGC, as applicable.

5.3.2.2 One-gallon (4-liter) quantity. Six 1-gallon (4-liter) cans of toluene shall be packed as specified in 5.3.2.1, except the pattern shall be two by three.

5.4 Marking. Shipments for civil agencies shall be marked in accordance with Fed. Std. No. 123. Shipments for military activities shall be marked in accordance with MIL-STD-129.

5.4.1 Container compliance markings. Each shipping container shall be marked in accordance with 49 CFR 171 to 179 and either ICAO-TDGA or IMO-IMDGC, as applicable.

5.4.2 Hazard class label. Each shipping container and pallet load shall be labeled in accordance with 49 CFR 171 to 179 and either ICAO-TDGA or IMO-IMDGC, as applicable.

5.4.3 Proper shipping name. Each shipping container and pallet load shall be marked with the proper shipping name in accordance with 49 CFR 171 to 179 and either ICAO-TDGA or IMO-IMDGC, as applicable.

5.4.4 Precautionary markings. Each unit and shipping container shall be marked or labeled, as applicable, in accordance with 29 CFR 1900.1200(f) to show the required precautionary information. Each outer container shall be marked to show the top of the container by use of an arrow and the word "UP".

5.5 Palletization. All shipments of containers of 5 gallons or less shall conform to load type VI in accordance with MIL-STD-147 and the pallet shall conform to type IV or V of MIL-P-15011. When specified, all other shipments of toluene shall be palletized in accordance with the applicable requirements of MIL-STD-147.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Toluene is intended for use as a solvent or thinner for organic coatings, various resins, and chlorinated rubber. It is also used as a diluent for some cellulosic lacquers and dopes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Unit quantity required
(c) Level of packing required (see 5.3)
(d) Palletization and preservation thereof, if required (see 5.5).

6.3 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent mailing addresses for submissions of data are listed in FED-STD-313.

6.4 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.6 Sampling and testing precautions. This specification requires inspection of chemical material which is potentially hazardous to personnel. This specification does not purport to address all of the safety problems associated with its use. It is the responsibility of the user of this specification to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Subject term (key word) listing.

cellulosic lacquers
chlorinated rubber
diluent
dopes

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organic coatings
resins
solvent
thinner

MILITARY INTERESTS:

Custodians:

Army - EA
Navy - SH
Air Force - 68

Review activities:

Army - AR, MD, MI
Navy - OS
DLA - GS
Other - DS

User activities:

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITIES:

EPA
GSA-FSS (9FTE-10)
VA-OSS

Preparing activity:

Army - EA

Project No. 6810-1240

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